Displacement Current

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Lecture Objective

- To examine how the introduction of displacement current produces a self-consistent set of equations that predict important electromagnetic phenomena like propagating waves.

Maxwell's Equations (but Not Quite)

- Maxwell's equations in differential form in the static case:
- Including Faraday's law:
- The three experimental pillars of electromagnetic field theory are now incorporated into the governing equations and yet …

$$\nabla \cdot \vec{D} = \rho_V$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{H} = \vec{J}$$

Question

- What are the three experimental pillars of electromagnetics?

  (Refer back to previous lectures as needed.)

Continuity Equation

- As we have seen, conservation of charge leads to the continuity equation:

$$\nabla \cdot \vec{J} + \frac{\partial \rho_V}{\partial t} = 0$$

Ampere's Law and the Continuity Equation (1 of 2)

- The differential form of Ampere's law in the static case is

$$\nabla \times \vec{H} = \vec{J}$$

- The continuity equation is

$$\nabla \cdot \vec{J} + \frac{\partial \rho_V}{\partial t} = 0$$

- In the time-varying case, Ampere's law in the above form is inconsistent with the continuity equation

$$\nabla \cdot \vec{J} = \nabla \cdot \left(\nabla \times \vec{H}\right) = 0$$

Ampere's Law and the Continuity Equation (2 of 2)

- To resolve this inconsistency, Maxwell modified Ampere's law to read:

$$\nabla \times \vec{H} = \vec{J}_c + \frac{\partial \vec{D}}{\partial t}$$

conduction current density

displacement current density

- In other words, Maxwell realized that a time-varying electric field behaves like a current to produce a magnetic field.

Maxwell's Equations

- Maxwell's equations in differential form:

$$\nabla \cdot \vec{D} = \rho_V$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

Ampere's Law and the Continuity Equation

- The new form of Ampere's law is consistent with the continuity equation as well as with the differential form of Gauss's law

$$\nabla \cdot \vec{J} + \frac{\partial}{\partial t}\left(\underbrace{\nabla \cdot \vec{D}}_{\rho_V}\right) = \nabla \cdot \left(\nabla \times \vec{H}\right) = 0$$

Displacement Current (1 of 2)

- Ampere's law can now be written as

$$\boxed{\nabla \times \overline{H} = \overline{J} + \overline{J}_d}$$

$$\overline{J}_d = \frac{\partial \overline{D}}{\partial t} = \text{displacement current density } (\text{A/m}^2)$$

Displacement Current (2 of 2)

- *Displacement current* is the type of current that flows between the plates of a capacitor.
- *Displacement current* is the mechanism which allows electromagnetic waves to propagate in a non-conducting medium.
- *Displacement current* is a consequence of the three experimental pillars of electromagnetics.

Conduction to Displacement Current Ratio (1 of 4)

- Consider a conducting medium characterized by conductivity σ and permittivity ε.
- The conduction current density is given by

$$\vec{J} = \sigma \vec{E}$$

- The displacement current density is given by

$$\vec{J}_d = \frac{\partial \vec{D}}{\partial t} = \text{displacement current density } (\text{A/m}^2)$$

Conduction to Displacement Current Ratio (2 of 4)

- Assume that the electric field is a sinusoidal function of time:

$$E = E_0 \cos \omega t$$

- Then,

$$J = \sigma E_0 \cos \omega t$$
$$J_d = -\omega \varepsilon E_0 \sin \omega t$$

Conduction to Displacement Current Ratio (3 of 4)

- We have

$$\left|J\right|_{\max} = \sigma E_0$$

$$\left|J_d\right|_{\max} = \omega \varepsilon E_0$$

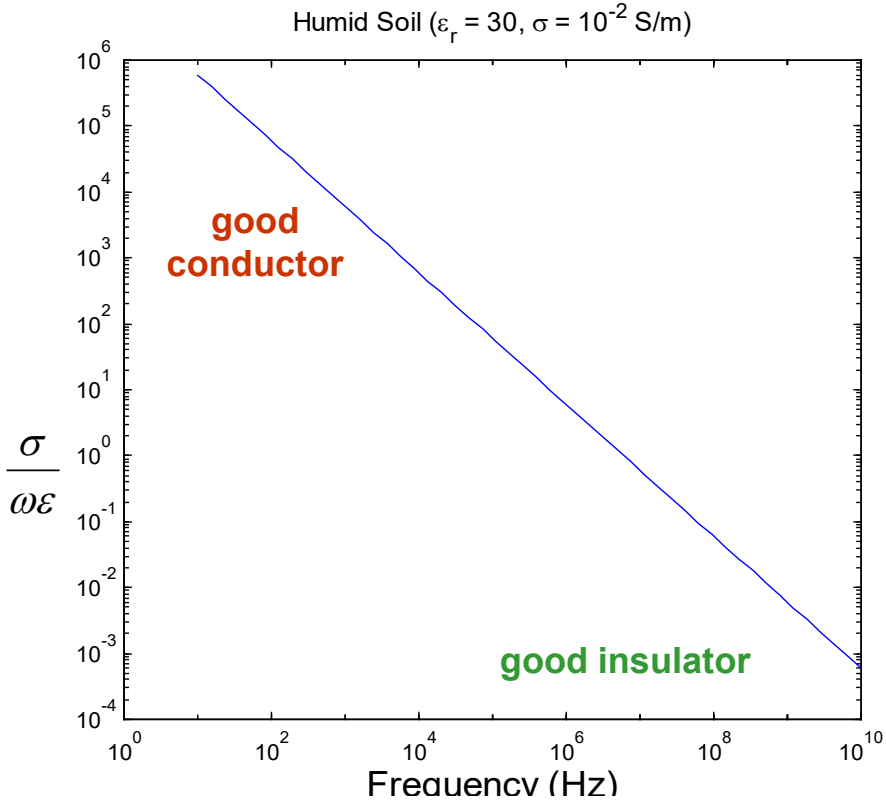
- Therefore

$$\boxed{\frac{\left|J\right|_{\max}}{\left|J_d\right|_{\max}} = \frac{\sigma}{\omega \varepsilon}}$$

Conduction to Displacement Current Ratio (4 of 4)

- The value of the quantity $\sigma/\omega\varepsilon$ at a specified frequency determines the properties of the medium at that given frequency.
- In a metallic conductor, the displacement current is negligible below optical frequencies.
- In free space (or other perfect dielectric), the conduction current is zero and only displacement current can exist.

Conduction to Displacement Current Ratio



Summary

- The introduction of the concept of displacement current allowed the development of a self-consistent set of equations that provide the classical description of all electromagnetic phenomena (Maxwell's equations).
- Displacement current is the mechanism which allows electromagnetic wave propagation.
- In a good insulator, displacement current is much greater than conduction current.
- In a good conductor, displacement current is negligible compared to conduction current.

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